

Quantifying the Maximum Recoverable Variance in Star Formation Rate from 1.4 GHz Radio Continuum Emission

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ABSTRACT

Radio continuum emission at ~ 1.4 GHz is widely employed as a dust-insensitive tracer of star formation rate (SFR) and forms a key science driver for current and forthcoming deep radio surveys. However, the extent to which radio luminosity alone permits accurate, deterministic SFR inference across different environments and luminosity regimes remains uncertain. We reassess the radio–SFR connection using deep multi-wavelength data for field galaxies from the VLA–COSMOS survey and cluster galaxies from the MeerKAT Galaxy Cluster Legacy Survey, spanning $0 < z \lesssim 6.5$. Infrared-derived SFRs are adopted as independent reference measurements. Using a suite of linear and non-linear regression models trained on radio-only inputs, we quantify the recoverable variance in SFR. Across all models, predictive performance saturates at $R^2 \sim 0.40$ – 0.45 in log-space, indicating that more than half of the intrinsic variance in SFR is not encoded in 1.4 GHz luminosity alone. The empirical radio–SFR relation has a slope significantly shallower than unity ($m \simeq 0.57$), implying luminosity-dependent biases if proportional calibrations are assumed. Masking independently identified AGN restores only a modest improvement, confirming that the variance ceiling is not driven by AGN contamination but reflects a broader physical decoupling between radio luminosity and star formation. Removing radio-excess sources does substantially tighten the observed relation. However, the physical nature of radio-excess sources cannot be unambiguously determined from radio data alone: these sources are morphologically distinct from both confirmed AGN and typical star-forming galaxies, with properties consistent with early-type or passive hosts, pointing to a secular rather than merger-driven channel. Excluding them from survey-scale analyses is therefore not physically justified without resolved multi-wavelength classification. These results suggest that the scatter in the radio–SFR relation is driven by a physically heterogeneous population whose properties cannot be recovered from radio luminosity alone. While radio emission remains a valuable tracer of star-forming activity, robust SFR inference in the MeerKAT and Square Kilometre Array era will require multi-wavelength or probabilistic approaches that account for intrinsic scatter and environmental modulation.

Keywords: Galaxies (573) – Galaxy Clusters (584) – Intracluster medium (858) – Star Formation (1569) – Radio Astronomy (1338)

1. INTRODUCTION

Measuring galaxy star formation rates (SFRs) across cosmic time is central to understanding how baryons cycle through galaxies and assemble stellar mass. Star formation diagnostics span ultraviolet, optical recombination lines, infrared (IR), and radio wavelengths, each sensitive to different physical timescales and systematic effects (Kennicutt 1998; Murphy et al. 2011). Among these, radio continuum emission near ~ 1.4 GHz has become particularly prominent because it is largely insensitive to dust extinction and can be observed efficiently over wide areas (e.g., Condon 1992; Murphy et al. 2011; Smolčić et al. 2017). Radio luminosity has been used as a quantitative SFR tracer across a

wide range of redshifts and environments (e.g., Seymour et al. 2008; Novak et al. 2017; Algera et al. 2020; Cochrane et al. 2023; Matthews et al. 2024), with a key requirement in that body of work being the separation of star-forming galaxies from AGN-dominated radio sources (e.g., Algera et al. 2020; Novak et al. 2017). This advantage is especially compelling in the era of deep, wide-field surveys with MeerKAT and the forthcoming Square Kilometre Array (SKA), where radio continuum will serve as a primary discovery space for star-forming galaxies across a broad range of redshifts and environments (e.g., Knowles et al. 2022).

The widespread use of radio emission as an SFR tracer is anchored in the far-infrared–radio correlation (FRC), which links dust-reprocessed emission from young massive stars to predominantly non-thermal synchrotron radiation produced by cosmic-ray electrons accelerated in supernova remnants (de Jong et al. 1985; Helou et al. 1985; Condon 1992). This empirical relation underpins commonly adopted radio–SFR calibrations that assume a monotonic, often approximately linear mapping between 1.4 GHz luminosity and star formation activity (Condon 1992; Bell 2003). However, radio luminosity depends not only on star formation but also on magnetic field strength, cosmic-ray transport, and energy-loss processes that vary with galaxy mass, structure, redshift, and environment (Bell 2003; Lacki et al. 2010; Murphy 2009; Schleicher & Beck 2016). In dense environments, additional non-thermal emission associated with intracluster processes may further complicate interpretation (e.g., Padovani et al. 2015; Croston et al. 2019). These factors introduce intrinsic scatter and potential departures from linearity in the radio–SFR relation.

AGN contamination presents an additional challenge. Radio emission from accretion-driven activity can mimic or exceed star-formation-powered synchrotron emission (e.g., Delvecchio et al. 2017, 2021). A common response is to remove “radio-excess” systems, typically defined by deviations from the mean infrared–radio relation (Delhaize et al. 2017; Smolčić et al. 2017). Doing so does substantially tighten the observed radio–SFR relation, which may appear to validate the exclusion. However, radio excess alone is not a clean AGN diagnostic: it can arise from enhanced magnetic fields, efficient cosmic-ray calorimetry, and time-dependent star formation histories without any AGN contribution (Lacki & Thompson 2010; Delhaize et al. 2017). More critically, independently masking confirmed AGN does not recover a tight radio–SFR relation; the variance ceiling persists regardless, demonstrating that AGN contamination is not the primary driver of the scatter. The physical nature of radio-excess sources is also poorly constrained without resolved morphological or spectroscopic information: these sources are systematically smoother and more centrally concentrated than matched non-radio-excess controls at fixed stellar mass, redshift, and SFR (Table 3), with properties consistent with passive early-type galaxies (Best et al. 2005; Croton et al. 2006; Hardcastle & Croston 2020) rather than with the disturbed merger-driven AGN that the conventional q_{IR} cut implicitly targets. In practice, especially for wide-area radio surveys lacking uniform multi-wavelength coverage, this ambiguity cannot be resolved, and excluding radio-excess sources imposes a selection that is neither physically justified nor uniformly applicable.

Beyond sample selection, a more fundamental issue concerns the informational content of radio luminosity itself. Much of the literature has treated the radio–SFR relation primarily as a calibration problem, measuring slopes and scatter under the assumption of a deterministic mapping (Condon 1992; Bell 2003; Delhaize et al. 2017; Novak et al. 2017). Here we instead pose an information-content question: *how much of the variance in SFR is recoverable from radio continuum observables alone?* If radio luminosity encodes sufficient information about star formation, flexible non-linear models should recover it; if instead performance saturates across model classes, this implies an intrinsic limit in the recoverable SFR information.

In this work, we directly quantify this recoverable variance using deep multi-wavelength data from the VLA–COSMOS 3 GHz Large Project (Smolčić et al. 2017) and cluster fields from the MeerKAT Galaxy Cluster Legacy Survey (MGCLS; Knowles et al. 2022), spanning environments from dense cluster cores to the field and extending to $z \lesssim 6.5$. Infrared-derived SFRs provide independent reference measurements: in COSMOS, SFRs are obtained via full IR SED fitting with explicit AGN components (Delvecchio et al. 2017; Berta et al. 2013), while in the cluster sample we adopt WISE-based mid-IR calibrations (Cluver et al. 2017; WISE Team 2020). We first establish the empirical radio–SFR relation and a simple linear correlation baseline (Section 4), then deploy a suite of linear and non-linear regression models trained on radio-only inputs to test whether any hidden predictive structure exists beyond simple proportional scalings. Convergence of linear and non-linear models at the same performance ceiling is itself a key result, demonstrating that the limitation is physical rather than a consequence of fitting methodology. Importantly, we retain radio-excess sources rather than excluding them via IR–radio ratio cuts, allowing us to evaluate radio-based SFR inference under realistic observational conditions.

Throughout this work, we assume a flat Λ CDM cosmology with parameters $H_0 = 67.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_M = 0.307$, and $\Omega_\Lambda = 0.693$, consistent with *Planck* 2018 results (Planck Collaboration et al. 2020). Luminosity distances are computed using `astropy.cosmology` (Astropy Collaboration et al. 2013, 2018).

The structure of this paper is as follows. In Section 2, we describe the construction of the field and cluster samples, the derivation of infrared-based SFRs, and our treatment of AGN and radio-excess sources. In Section 3, we outline the machine learning framework, define the concept of recoverable variance, and detail the training and evaluation strategy used to quantify the informational content of radio luminosity. Section 4 presents the empirical radio–SFR relation, the predictive performance of radio-only models, and tests of environmental dependence in cluster environments. In Section 5, we interpret the observed variance ceiling in terms of cosmic-ray physics, magnetic field variation, star formation history, and environmental effects, and discuss implications for current and forthcoming radio surveys. Finally, Section 6 summarizes our principal findings and outlines consequences for radio-based SFR inference in the MeerKAT and SKA era.

2. DATA

Our analysis combines deep radio continuum observations with independent infrared-based star formation rate (SFR) measurements across a wide range of environments and redshifts. We draw on two complementary datasets: the VLA–COSMOS 3 GHz Large Project, which provides a well-characterized field sample with extensive multi-wavelength coverage, and the MeerKAT Galaxy Cluster Legacy Survey (MGCLS), which enables a direct assessment of environmental effects in dense cluster regions. Together, these datasets allow us to evaluate the reliability of radio luminosity as a star formation tracer under both ideal and realistic observational conditions.

A central requirement of this work is that SFR estimates be derived independently of the radio data used for prediction. In the COSMOS field, this is achieved via full infrared SED fitting incorporating far-infrared data and explicit AGN components, yielding robust, dust-unbiased SFRs (Delvecchio et al. 2017; Berta et al. 2013). In the cluster sample, where homogeneous far-infrared coverage is unavailable, we adopt mid-infrared SFR estimates based on *WISE* photometry and empirical calibrations (Cluver et al. 2017). Although these two approaches differ in methodology, both provide physically motivated benchmarks against which the information content of radio luminosity can be tested.

Radio luminosities throughout this work are quoted as $\log_{10}(L_{1.4 \text{ GHz}} [\text{W Hz}^{-1}])$. Where flux densities are required for comparison with observed catalogues, these are given in units of mJy and are clearly distinguished from luminosities in the text and figures.

Radio luminosities are computed assuming a power-law radio spectrum, $S_\nu \propto \nu^\alpha$, with spectral index $\alpha = -0.7$, typical of synchrotron emission from star-forming galaxies (Condon 1992). While individual sources may deviate from this assumption, we show that our qualitative conclusions are insensitive to reasonable variations in spectral index and redshift uncertainties.

In the following subsections, we describe the construction of the COSMOS and cluster samples, the environmental characterisation of cluster galaxies (including cluster properties and source counts), the derivation of infrared-based SFRs for the cluster sample, and our treatment of AGN and radio-excess sources.

2.1. VLA–COSMOS 3 GHz Large Project

The VLA–COSMOS 3 GHz Large Project provides deep radio continuum imaging over 2 deg^2 with a median rms sensitivity of $\sim 2.3 \mu\text{Jy beam}^{-1}$ and an angular resolution of $0''.75$ (Smolčić et al. 2017). The survey detects $\sim 10^4$ radio sources above 5σ significance, spanning star-forming galaxies and AGN across a broad redshift range.

We select galaxies with reliable 3 GHz detections above 5σ (corresponding to a flux density limit of $\sim 11.5 \mu\text{Jy}$ at the median rms), measured 1.4 GHz flux densities, redshift estimates, and infrared-derived SFRs. Multi-wavelength counterparts are obtained via likelihood-ratio matching (Sutherland & Saunders 1992) to the COSMOS2015 catalogue (Laigle et al. 2016), providing photometric redshifts, stellar masses, and ancillary data. This sample forms the basis for training and validation of our machine learning models.

Infrared luminosities and SFRs are taken from Delvecchio et al. (2017), who derived total infrared (8–1000 μm) luminosities using the SED3FIT SED-fitting framework (Berta et al. 2013). This approach simultaneously models stellar emission, dust-reprocessed star formation, and AGN components using *Spitzer* and *Herschel* photometry, ensuring that the inferred SFRs are not contaminated by AGN-heated dust (Fritz et al. 2006; Feltre et al. 2012). These IR-based SFRs provide a robust, independent reference against which radio-based predictions can be evaluated.

2.2. MeerKAT Galaxy Cluster Legacy Survey

To assess environmental effects, we apply our analysis to 24 galaxy clusters drawn from the MeerKAT Galaxy Cluster Legacy Survey (Knowles et al. 2022). MGCLS provides 1.28 GHz radio continuum imaging of 115 clusters over $> 20 \text{ deg}^2$, with typical rms sensitivities of $2\text{--}5 \mu\text{Jy beam}^{-1}$ and angular resolution of $\sim 8''$.

Compact radio sources are identified using PyBDSF (Mohan & Rafferty 2015). Optical and infrared counterparts are obtained via likelihood-ratio matching (Sutherland & Saunders 1992) to DES (Dark Energy Survey Collaboration et al. 2016) and *WISE* (WISE Team 2020) catalogues. The resulting radio-selected samples include star-forming galaxies, AGN, and systems affected by cluster-specific processes such as ram-pressure stripping, shocks, and diffuse synchrotron emission (e.g., Dressler 1980; Cuciti et al. 2021).

MGCLS and COSMOS-VLA enable a direct comparison of the radio–SFR relation inside and outside cluster environments, where additional non-star-forming radio emission from relativistic electrons in the intracluster medium may further complicate the interpretation of radio luminosity (e.g., Croston et al. 2019).

2.3. Cluster Properties, Environmental Characterisation, and Source Counts

For each cluster, we adopt published redshifts and characteristic radii (R_{500}) from X-ray, lensing, and dynamical analyses (e.g., Piffaretti et al. 2011; Cuciti et al. 2021; Mantz et al. 2016; Umetsu et al. 2022). Here, R_{500} denotes the radius within which the mean enclosed mass density is 500 times the critical density of the Universe at the cluster redshift, and is commonly used as a physically motivated scale that approximates the virialized region accessible to X-ray observations. Where necessary, we approximate $R_{500} \simeq R_{200}/1.5$ ($R_{500}/R_{200} \approx 0.65$), consistent with Navarro, Frenk, & White (NFW; 1997) halo profiles and typical cluster concentrations of $c_{200} \sim 4\text{--}5$. Projected cluster-centric distances are computed for all radio sources, allowing the sample to be divided into galaxies located within R_{500} (cluster cores) and those at larger radii. This separation enables an explicit test of whether dense environments degrade the reliability of radio luminosity as a tracer of star formation.

To characterize cluster environments, we incorporated published R_{500} and mass parameters from:

- The *Meta-catalogue of X-ray Detected Clusters of Galaxies (MCXC)* (Piffaretti et al. 2011);
- Mass-selected cluster samples and radio halo studies (Cuciti et al. 2021);
- Detailed analyses of individual systems (e.g., Abell 133, Abell 194, Abell 209, El Gordo; Fujita et al. 2013; Thuruthipilly & Junais 2024; Cuciti et al. 2021; Zitrin et al. 2013);
- Environmental studies of cluster galaxy evolution (Venkatapathy et al. 2017);
- Additional dynamical, lensing and X-ray reconstructions (Mantz et al. 2016; Urdampilleta et al. 2021; Umetsu et al. 2022).

Table 1 summarizes the MGCLS cluster fields used in this work, including pointing coordinates, redshifts, adopted R_{500} values, and the corresponding literature references.

For completeness, Table 2 summarizes the number of compact radio sources detected in each MGCLS cluster field, the number of those sources with reliable *WISE* infrared counterparts, and the final matched sample used in the machine-learning analysis. Infrared counterparts are identified only at the positions of radio-detected sources; consequently, all infrared sources considered in this work are associated with radio detections by construction. The radio+*WISE* sample therefore represents a subset of the radio-selected population, rather than an independent parent sample. These source counts are provided to document the statistical weight of each cluster field and to facilitate reproducibility of the analysis.

2.4. Infrared-based Star Formation Rates

For the cluster sample, star formation rates are estimated using *WISE* *W4* ($22 \mu\text{m}$) photometry and the empirical calibration of Cluver et al. (2017). Observed *W4* magnitudes are converted to flux densities using the *WISE* zero-point of $F_{0,W4} = 8.283 \text{ Jy}$ (Wright et al. 2010). The flux density is then converted to a monochromatic luminosity using the luminosity distance at the cluster redshift.

To account for bandpass shifting with redshift, we apply mid-infrared *k*-corrections derived from fits to the observed *W1*–*W4* photometry using a simple two-component SED model, consisting of a Rayleigh–Jeans stellar component

Table 1. Galaxy Cluster Properties

Cluster	RA (deg)	Dec (deg)	Redshift	R_{500} (Mpc)	No. of Sources	R_{500} Ref.
Abell 22	5.1608	-25.7220	0.142	1.0560	3417	Arnaud et al. (2010)
Abell 85	10.4529	-9.3180	0.055	1.2000	2470	Venkatapathy et al. (2017)
Abell 133	15.6879	-21.8800	0.057	1.0020	2930	Fujita et al. (2013)
Abell 194	21.4458	-1.3731	0.018	0.6549	907	Thuruthipilly & Junais (2024)
Abell 209	22.9896	-13.5764	0.210	0.8867	2481	Cuciti et al. (2021)
Abell 2485	342.1371	-16.1062	0.247	1.0622	3627	Piffaretti et al. (2011)
Abell 2597	351.3321	-12.1244	0.085	1.1066	1011	Piffaretti et al. (2011)
Abell 2645	355.3200	-2.0975	0.251	1.0897	2009	Piffaretti et al. (2011)
Abell 2667	357.9196	-26.0836	0.230	1.3031	4001	Piffaretti et al. (2011)
Abell 2744	3.5671	-30.3830	0.308	1.2359	3048	Piffaretti et al. (2011)
Abell 2751	4.0580	-31.3885	0.107	0.7431	2111	Piffaretti et al. (2011)
Abell 2811	10.5368	-28.5358	0.108	1.0355	773	Piffaretti et al. (2011)
Abell 2895	19.6454	-26.9731	0.227	1.0896	3383	Piffaretti et al. (2011)
Abell 3365	87.0500	-21.9350	0.093	0.8100	1067	Urdampilleta et al. (2021)
Abell 3376	90.4256	-39.9851	0.046	0.8726	2421	Piffaretti et al. (2011)
Abell 3558	201.9783	-31.4922	0.048	1.1010	2645	Piffaretti et al. (2011)
Abell 3562	202.7833	-31.6713	0.049	0.9265	2547	Piffaretti et al. (2011)
Abell 3667	303.1403	-56.8406	0.056	1.1990	1176	Piffaretti et al. (2011)
Abell 370	39.9604	-1.5856	0.375	1.0933	1407	Umetsu et al. (2022)
Abell 4038	356.8796	-28.2028	0.028	0.8863	3593	Piffaretti et al. (2011)
Abell 521	73.5358	-10.2442	0.253	1.2400	2423	Cuciti et al. (2021)
Abell 545	86.7571	-25.6164	0.154	1.3500	2548	Mantz et al. (2016)
El Gordo	15.7188	-49.2495	0.870	1.1500	3127	Zitrin et al. (2013)
J0014.3-6604	3.5767	-66.0775	0.155	1.0117	4459	Piffaretti et al. (2011)

Notes: (1) Cluster names listed alphabetically and by ascending Abell number, with non-Abell clusters listed last; (2) and (3) MeerKAT pointing coordinates: J2000 RA and Dec (Knowles et al. 2022); (4) Cluster redshift (Knowles et al. 2022); (5) R_{500} ; (6) Number of sources in the cluster field; (7) R_{500} reference.

and a power-law dust component. The best-fitting SED is evaluated at an observed wavelength of $22(1+z)\mu\text{m}$, corresponding to rest-frame $22\mu\text{m}$ emission. While monochromatic mid-infrared estimators are sensitive to dust temperature, metallicity, and star formation timescales, they provide a practical and widely used SFR proxy in the absence of homogeneous far-infrared coverage (Cluver et al. 2017).

Although COSMOS and cluster SFRs are derived using different methodologies, these differences primarily introduce additional scatter and modest systematic offsets rather than order-of-magnitude discrepancies. Since our analysis focuses on correlations, variance, and predictive limits rather than absolute SFR normalization, these differences do not qualitatively affect our conclusions.

2.5. Infrared–Radio Ratio and Radio-excess Sources

The relative strength of infrared and radio emission is commonly quantified using the infrared–radio ratio q (Helou et al. 1985), defined as

$$q_{\text{IR}} \equiv \log_{10} \left(\frac{L_{\text{IR}}}{3.75 \times 10^{12} \text{ W}} \right) - \log_{10} \left(\frac{L_{1.4 \text{ GHz}}}{\text{W Hz}^{-1}} \right), \quad (1)$$

where L_{IR} is the total infrared luminosity integrated over 8–1000 μm and $L_{1.4 \text{ GHz}}$ is the monochromatic radio luminosity at 1.4 GHz in units of W Hz^{-1} .

Star-forming galaxies exhibit a characteristic distribution in q_{IR} with intrinsic scatter of ~ 0.2 – 0.3 dex (Yun et al. 2001; Bell 2003). Systems with lower q_{IR} values are often labelled as radio-excess sources and frequently interpreted as AGN-dominated (Delvecchio et al. 2021; Smolčić et al. 2017).

However, radio excess does not uniquely imply AGN activity. Intrinsic scatter in the IR–radio correlation, compact starbursts with enhanced magnetic fields, efficient cosmic-ray calorimetry, and secondary production of relativistic

Table 2. Source counts per cluster. Columns list the total number of radio sources identified in the MGCLS compact source catalogue (N_{radio}), the number of radio sources with reliable WISE infrared detections (N_{WISE}), and the final matched sample used in the machine-learning analysis ($N_{\text{radio+WISE}}$).

Cluster	N_{radio}	N_{WISE}	$N_{\text{radio+WISE}}$
Abell 133	5040	2930	2930
Abell 194	1422	907	907
Abell 209	4456	2481	2481
Abell 22	6119	3417	3417
Abell 2485	7170	3627	3627
Abell 2597	1661	1011	1011
Abell 2645	3488	2009	2009
Abell 2667	7307	4001	4001
Abell 2744	5224	3048	3048
Abell 2751	3610	2111	2111
Abell 2811	1256	773	773
Abell 2895	6059	3383	3383
Abell 3365	2050	1067	1067
Abell 3376	4370	2422	2422
Abell 3558	5644	2645	2645
Abell 3562	5442	2547	2547
Abell 3667	2099	1176	1176
Abell 370	2261	1405	1405
Abell 4038	6197	3593	3593
Abell 521	4342	2423	2423
Abell 545	5039	2548	2548
Abell 85	4629	2470	2470
El Gordo	5558	3127	3127
J0014.3–6604	8456	4459	4459

All WISE sources are associated with radio detections by construction; infrared counterparts and star formation rates were derived only for radio-selected sources. Consequently, $N_{\text{WISE}} = N_{\text{radio+WISE}}$ for all clusters.

electrons and positrons in dense gas can all increase synchrotron emissivity per unit SFR (e.g., [Thompson et al. 2007](#); [Murphy 2009](#); [Yoast-Hull et al. 2016](#)). In addition, time-dependent star formation histories can temporarily depress infrared emission relative to radio emission, producing offsets of up to $\lesssim 0.5$ dex below the mean IR–radio relation without any AGN contribution ([Bressan et al. 2002](#); [Delhaize et al. 2017](#)). Only $\sim 0.7\%$ of the radio-excess sources in our sample lie beyond 0.5 dex of the non-AGN mean.

We find that removing radio-excess sources does substantially restore a tighter radio–SFR relation (Table 4). Applying the standard 2σ threshold ($q_{\text{IR}} < \bar{q} - 2\sigma_q$; [Delvecchio et al. 2021](#); [Smolčić et al. 2017](#)) removes 378 sources (4.9% of the sample) and raises the Pearson correlation from $r = 0.652$ to $r = 0.789$, the best-fit slope from $m = 0.574$ to $m = 0.724$, and the best-performing model R^2 from 0.459 to 0.633. A stricter 1σ cut (removing 14% of the sample) raises these further to $r = 0.876$, $m = 0.811$, and $R^2 = 0.739$. The relation does tighten, and tightens substantially.

We nevertheless retain radio-excess sources in our primary analysis for two reasons, which together constitute a physical and methodological argument against their routine exclusion.

First, the improvement is an artefact of what radio-excess sources *are*, not evidence of what they *are not*. Radio-excess is defined as elevated radio emission relative to infrared emission, that is, these sources lie off the radio–SFR relation by construction. Removing sources that are offset from a relation will always tighten that relation, regardless of why they are offset. The standard literature practice of applying a q_{IR} cut (e.g., [Delvecchio et al. 2017](#); [Smolčić et al. 2017](#); [Delhaize et al. 2017](#); [Novak et al. 2017](#)) therefore does not demonstrate that the resulting sample is physically

clean; it demonstrates that the outliers have been removed. The degree of tightening is proportional to how far the excluded sources lie from the relation, not to the physical validity of the exclusion criterion (Condon 1992).

Second, and more decisively, the physical nature of these sources does not support their blanket classification as AGN. To characterise the morphological properties of radio-excess hosts, we use publicly available *Hubble Space Telescope* (HST) ACS F814W imaging from the COSMOS survey (Scoville et al. 2007), which provides $\sim 0''.09$ resolution I-band photometry across the full VLA–COSMOS footprint. Non-parametric morphological statistics, specifically the Gini coefficient (G ; Lotz et al. 2004), concentration index (C ; Conselice 2003), and rotational asymmetry (A ; Conselice 2003), are computed for each source using the STATMORPH package (Rodríguez-Gómez et al. 2019), which performs pixel-level measurements on background-subtracted, PSF-corrected cutouts. All measurements use a signal-to-noise threshold of $S/N \geq 2.5$ per pixel and a Petrosian elliptical aperture to define the galaxy extent. As an independent check on the non-parametric statistics, we cross-match the radio-excess and control samples against the Zurich Estimator of Structural Types (ZEST) catalogue (Scarlata et al. 2007), which provides visual and PCA-based morphological classifications (early-type, disc, and irregular) for $\sim 56\,000$ COSMOS galaxies from the same HST/ACS F814W imaging. The ZEST early-type fraction (Type 1) among radio-excess hosts is significantly higher than among matched controls, providing independent corroboration that these sources preferentially occupy morphologically undisturbed, bulge-dominated hosts rather than the disturbed or disc-dominated systems typical of merger-triggered AGN.

To isolate the effect of radio excess from confounding variables, each radio-excess source ($N_{\text{rx}} = 1003$) is matched one-to-one without replacement to a unique non-radio-excess control in standardised ($\log M_*$, z , $\log L_{\text{TIR}}$) space using nearest-neighbour matching with a maximum tolerance of 0.1σ per dimension. This ensures that any morphological difference between the two populations cannot be attributed to differences in stellar mass, redshift, or infrared luminosity. We additionally separate the radio-excess population by spectral index α_{radio} , computed between the 1.4 and 3 GHz flux densities, defining flat-spectrum sources as those with $\alpha_{\text{radio}} \geq -0.5$ ($N_{\text{rx}} = 139$) and steep-spectrum sources as those with $\alpha_{\text{radio}} < -0.5$ ($N_{\text{rx}} = 864$).

Our morphological characterisation (Table 3) shows that, after this matching, radio-excess sources show significantly higher Gini coefficients ($d = +0.22$; 95% CI $[+0.13, +0.31]$) and concentration indices ($d = +0.18$; $[+0.10, +0.27]$) than matched non-radio-excess controls, and are simultaneously smoother ($d_A = -0.14$; $[-0.23, -0.05]$). For the flat-spectrum sub-sample, the sources most likely to harbour compact, self-absorbed radio emission, the asymmetry suppression is extreme ($d = -0.87$; 95% CI $[-1.12, -0.64]$; $p = 8 \times 10^{-13}$): their median asymmetry is barely half that of individually matched controls at fixed mass, redshift, and SFR. These results are consistent with the ZEST classification of radio-excess hosts as preferentially early-type (Scarlata et al. 2007). These are not the morphologically disturbed, merger-triggered AGN hosts that dominate the radio-loud AGN literature (Holt et al. 2008; Hardcastle & Croston 2020; O’Dea & Saikia 2021). They are compact, smooth, and centrally concentrated, morphological properties consistent with passive early-type galaxies or systems with low-luminosity AGN sustained by secular gas supply rather than major mergers (Best et al. 2005; Croton et al. 2006; Hardcastle & Croston 2020). In the absence of resolved morphological or spectroscopic classification for all sources, classifying them as AGN contaminants and discarding them is not warranted. Retaining them allows us to evaluate the reliability of radio luminosity as a star formation tracer under the realistic and unavoidable conditions present in wide-field radio surveys.

2.6. AGN Identification

AGN in the COSMOS sample are identified following the multi-wavelength criteria described by Smolčić et al. (2017). X-ray AGN are selected using an intrinsic luminosity threshold of $L_X \geq 10^{42} \text{ erg s}^{-1}$ in the 0.5–8 keV band (Szokoly et al. 2004), a regime where X-ray emission from star formation is negligible (Symeonidis et al. 2014). Additional AGN are identified via mid-infrared colour diagnostics (Donley et al. 2012) and SED fitting that explicitly includes an AGN component (Delvecchio et al. 2014).

We do not attempt to identify radio-loud AGN in the cluster sample. Uniform deep X-ray and far-infrared data required for robust AGN classification are unavailable across all cluster fields, and radio luminosity alone does not provide an unambiguous diagnostic of AGN activity (Padovani et al. 2015; Hardcastle et al. 2025). While extremely radio-luminous sources are likely AGN-dominated, defining a physically motivated and redshift-independent luminosity threshold is non-trivial, particularly given the wide range of galaxy masses, environments, and potential contributions from star-formation-driven processes.

Crucially, however, the inability to classify cluster AGN does not affect our conclusions. In the COSMOS field, where robust multi-wavelength AGN identification is available, masking confirmed AGN does not recover a tight radio–SFR

Table 3. Morphological properties of radio-excess hosts relative to matched non-radio-excess controls, measured from HST/ACS F814W imaging (Scoville et al. 2007) using STATMORPH (Rodríguez-Gomez et al. 2019), with independent corroboration from the ZEST morphological catalogue (Scarlata et al. 2007). Each radio-excess source is matched one-to-one to a unique non-radio-excess control in standardised ($\log M_*$, z , $\log L_{\text{TIR}}$) space. Cohen’s d quantifies the standardised mean difference (positive = radio-excess hosts have larger values); 95% confidence intervals are from 5000 bootstrap resamples. KS p -values test the null hypothesis that the two distributions are drawn from the same parent population. All three effects are statistically robust and well-separated from zero. The flat-spectrum sub-sample ($\alpha_{\text{radio}} \geq -0.5$; $N_{\text{rx}} = 139$) shows extreme asymmetry suppression, consistent with morphologically undisturbed early-type hosts.

Sub-sample	Morphology metric	Cohen’s d	95% CI	KS p
3*Full matched sample ($N_{\text{rx}} = 1003$)	Gini coefficient	+0.22	[+0.13, +0.31]	4.6×10^{-6}
	Concentration	+0.18	[+0.10, +0.27]	6.2×10^{-4}
	Asymmetry	−0.14	[−0.23, −0.05]	3.6×10^{-4}
3*Steep-spectrum sub-sample ($\alpha_{\text{radio}} < -0.5$; $N_{\text{rx}} = 864$)	Gini coefficient	+0.19	[+0.10, +0.29]	3.1×10^{-4}
	Concentration	+0.16	[+0.07, +0.26]	1.2×10^{-3}
	Asymmetry	−0.12	[−0.22, −0.03]	1.1×10^{-2}
3*Flat-spectrum sub-sample ($\alpha_{\text{radio}} \geq -0.5$; $N_{\text{rx}} = 139$)	Gini coefficient	+0.28	[+0.05, +0.53]	1.5×10^{-2}
	Concentration	+0.35	[+0.11, +0.60]	1.0×10^{-2}
	Asymmetry	−0.87	[−1.12, −0.64]	8.0×10^{-13}

Notes: Positive d indicates radio-excess hosts are more compact or concentrated; negative d indicates they are smoother. The flat-spectrum asymmetry suppression ($d = -0.87$) is the dominant structural signal: the median asymmetry of flat-spectrum radio-excess hosts is barely half that of their individually matched controls, robust to spectral-index significance thresholds up to $2\sigma_\alpha$ ($d = -0.57$, $p = 0.003$). These morphological properties are consistent with passive early-type galaxies or systems with low-luminosity AGN fuelled through secular processes, rather than merger-triggered AGN or compact starbursts (Best et al. 2005; Croton et al. 2006; van der Wel et al. 2014; Hardcastle & Croston 2020), and are corroborated by the higher ZEST early-type fraction (Type 1; Scarlata et al. 2007) among radio-excess hosts relative to matched controls.

Table 4. Effect of radio-excess source removal on the radio–SFR relation and predictive performance. Radio-excess sources are defined as $q_{\text{IR}} < \bar{q} - N\sigma_q$ where $\bar{q} = 2.076$ and $\sigma_q = 0.606$ for the VLA–COSMOS sample. R^2 values are test-set scores in $\log_{10}(\text{SFR})$ space. CI half-widths are means across luminosity bins (see Table 6). The improvement in r , slope, and R^2 upon removal reflects the fact that radio-excess sources are offset from the radio–SFR relation by definition; removing outliers necessarily tightens any relation regardless of the physical cause of the offset. The retained AGN (identified via X-ray, MIR, and SED criteria) do not drive the improvement, confirming that the radio-excess and AGN populations are not equivalent.

	Full sample	Non-excess (2σ)	Non-excess (1σ)
N (sources)	7729	7351	6650
Fraction of full sample	100%	95.1%	86.0%
Best-fit slope m	0.574	0.724	0.811
Pearson r	0.652	0.789	0.876
Mean 68% CI half-width (dex)	0.123	0.106	0.095
Mean 95% CI half-width (dex)	0.420	0.365	0.297
<i>Test-set R^2 by model</i>			
Random Forest	0.198	0.408	0.584
Linear Regression	0.332	0.608	0.730
Support Vector Regression	0.459	0.633	0.739
k -Nearest Neighbors	0.355	0.510	0.663
Gradient Boosting	0.446	0.604	0.725
Neural Network (MLP)	0.438	0.616	0.733

Notes: The 2σ cut corresponds to $q_{\text{IR}} < 0.863$; the 1σ cut to $q_{\text{IR}} < 1.469$. The improvement in R^2 upon removal (e.g. from 0.459 to 0.633 for SVR) reflects the exclusion of sources that are offset from the mean relation, not a demonstration that those sources are AGN. Even after removal, AGN identified independently via X-ray and MIR criteria remain in the sample, yet the relation tightens substantially: the radio-excess and AGN populations are therefore not the same population.

relation: predictive performance remains limited at $R^2 \lesssim 0.45$ even after AGN removal (Section 4.2). The variance ceiling is therefore not driven by AGN contamination. Applying the same models to the cluster sample without AGN removal is therefore internally consistent with our COSMOS result, and provides a realistic characterisation of what radio-based SFR inference can achieve under typical wide-field survey conditions.

Importantly, masking independently identified AGN in the COSMOS field does not eliminate the variance ceiling: predictive performance remains limited at $R^2 \lesssim 0.45$ even after AGN removal. The radio–SFR relation is therefore not solely a problem of obvious AGN contamination, but reflects a broader physical decoupling between radio luminosity and star formation (Lacki et al. 2010; Murphy 2009). Retaining all sources in the cluster sample therefore provides a more realistic and internally consistent test of whether radio luminosity alone can recover star formation rates.

3. METHODS

The aim of this work is to quantify the maximum amount of information about star formation that can be extracted from radio continuum emission alone. To address this, we adopt a machine learning (ML) framework that imposes minimal assumptions on the functional form of the radio–SFR relation. If radio luminosity encodes sufficient information about star formation, flexible non-linear models should be able to recover it. Conversely, systematic failure across a broad class of models indicates that the limitation is physical rather than methodological.

All models are trained using the VLA–COSMOS field sample, where high-quality infrared-derived SFRs and redshift information are available, and are evaluated on both held-out COSMOS data and independent cluster samples from the MGCLS. Throughout, infrared-based SFRs are treated as reference quantities, and radio luminosities as the predictors whose informational content is being tested.

3.1. Input Features and Target Variable

The target variable for all supervised learning experiments is the base-10 logarithm of the star formation rate, $\log_{10}(\text{SFR}/\text{M}_{\odot} \text{ yr}^{-1})$, where taking the logarithm reduces the dynamic range of the data and mitigates heteroscedastic scatter (Bell 2003; Delhaize et al. 2017).

The primary input feature is the rest-frame 1.4 GHz radio luminosity, $\log_{10}(L_{1.4 \text{ GHz}} [\text{W Hz}^{-1}])$. In extended feature sets, we additionally include 3 GHz radio luminosity and redshift to test whether modest increases in radio-related information improve predictive performance. No infrared information is included in any radio-only model.

This conservative feature selection is intentional. Our aim is to quantify the extent to which radio luminosity alone can act as a stand-alone SFR proxy — the implicit assumption behind radio-only SFR catalogues (Condon 1992; Bell 2003; Murphy et al. 2011). Including auxiliary properties such as stellar mass or morphology would change the scientific question from “what does radio encode?” to “what can a multivariate model achieve?”, which is a separate problem. Extended models include at most redshift and a second radio band to test whether modest increases in radio-domain information alter the recoverable variance.

3.2. Machine Learning Models

We implement six supervised regression algorithms using the SCIKIT-LEARN library (Pedregosa et al. 2011), selected to span a wide range of model complexity and inductive bias:

- **Linear Regression**, which tests the assumption of a single global linear relation in log–log space.
- **Support Vector Regression (SVR)** with a radial basis function kernel, which captures smooth non-linear dependencies.
- **k -Nearest Neighbours (kNN)** regression, which probes whether local similarity in radio luminosity corresponds to similar SFRs.
- **Random Forest Regression**, which models hierarchical, non-linear feature interactions and is robust to outliers.
- **Gradient Boosting Regression**, which builds additive corrections to systematic residuals.
- **Multilayer Perceptron (MLP)** neural network, consisting of two hidden layers with 64 and 32 neurons, trained to minimize the mean squared error in log-space.

This ensemble allows us to test whether predictive failure is specific to a particular modelling choice or persists even under highly flexible, non-parametric regression.

3.3. Training and Evaluation

Models are trained using an 80/20 train–test split with a fixed random seed to ensure reproducibility. The split is performed after random shuffling, and we verify that both subsets span the full range of redshift and radio luminosity present in the parent sample. This prevents artificial performance inflation or degradation arising from uneven sampling of parameter space.

All models are trained to predict $\log_{10}(\text{SFR})$ and are first evaluated on the held-out COSMOS test set. Model performance is quantified primarily using the coefficient of determination,

$$R^2 = 1 - \frac{\sum (y_{\text{true}} - y_{\text{pred}})^2}{\sum (y_{\text{true}} - \bar{y}_{\text{true}})^2}, \quad (2)$$

computed in log-space. Physically, R^2 measures the fraction of the variance in star formation rate that can be explained by the input radio features alone. All R^2 values reported in this paper are test-set scores unless otherwise stated. For five of the six models (Linear Regression, SVR, kNN, Gradient Boosting, and MLP), training-set R^2 values are within 0.02–0.05 of their test-set counterparts, confirming minimal overfitting. The exception is Random Forest, which achieves $R^2_{\text{train}} = 0.897$ but $R^2_{\text{test}} = 0.198$, indicating severe overfitting when trained on 1.4 GHz luminosity alone; for this reason, Random Forest predictions are treated with caution and the ensemble ceiling is better characterised by the other five models. The consistent performance ceiling across the five well-generalising models confirms that the limit is a genuine informational constraint rather than a model generalisation failure.

To provide a physically meaningful performance baseline, we bootstrap the expected R^2 under a perfect mapping: we draw radio luminosities from the observed distribution, apply the best-fitting linear relation, add Gaussian scatter of amplitude σ_{obs} (the residual standard deviation around the best-fit line, which characterises the observational and intrinsic uncertainty in our sample), and fit a linear model to each realisation. The resulting distribution of R^2 values, centred near $R^2_{\text{boot}} \approx 0.43$ (90% CI [0.41, 0.44]) for the observed scatter of $\sigma_{\text{obs}} = 0.51$ dex, represents the performance achievable by a perfect, noise-limited linear mapping and serves as our fair upper-bound comparison. This value is considerably lower than unity because the intrinsic scatter in the radio–SFR relation ($\sigma_{\text{obs}} = 0.51$ dex) is large: even a perfect proportional mapping cannot achieve a high R^2 when the residual noise is this substantial (Yun et al. 2001; Bell 2003). All reported model R^2 values are interpreted relative to this bootstrap baseline.

To retain physical interpretability, we also compute mean and median absolute errors in linear SFR space, which reflect the typical magnitude of misestimates expected when inferring SFRs from radio data alone.

3.4. AGN and Radio-excess Sources

AGN identified in the COSMOS sample via X-ray, mid-infrared, or SED-based diagnostics are retained in the primary analysis, though results are also examined separately for AGN and non-AGN subsamples. We do not exclude radio-excess sources using infrared–radio ratio cuts.

This choice is deliberate. We find that removing radio-excess sources does tighten the observed relation, but this tightening reflects a selection effect rather than a physical improvement in the radio–SFR mapping: the excluded systems are a morphologically and physically heterogeneous population that cannot be uniformly classified as AGN from radio data alone. As detailed in Section 2.5, radio-excess hosts are compact, smooth, and concentrated, properties associated with early-type galaxies (van der Wel et al. 2014), rather than with the disturbed or star-forming morphologies expected if the excess arose from merger-driven AGN triggering (Holt et al. 2008; O’Dea & Saikia 2021). Their exclusion therefore removes a physically real component of the galaxy population without robust justification, and would not be possible in a uniform way across the heterogeneous multi-wavelength coverage of wide-field radio surveys.

By retaining radio-excess sources, we evaluate the reliability of radio continuum emission as a star formation tracer under realistic observational conditions directly relevant to MeerKAT and forthcoming SKA surveys.

3.5. Application to Cluster Environments

Models trained on the COSMOS field sample are applied without retraining to the MGCLS cluster data. This transfer tests whether any radio–SFR relation learned in the field generalizes to dense environments, where additional non-star-forming radio emission from AGN activity and relativistic electrons in the intra-cluster medium is expected (Croston et al. 2019; Cuciti et al. 2021).

For cluster sources, rest-frame radio luminosities are computed assuming the cluster redshift. Sources are further divided by projected cluster-centric distance relative to R_{500} to assess environmental dependence. This approach allows us to test not only whether radio luminosity predicts star formation, but whether any predictive structure is robust to environmental complexity. Where flux densities are required for direct comparison with observations, luminosities are converted using $S_{\nu_{\text{obs}}} = (L_{\nu}/4\pi D_L^2)(1+z)^{1-\alpha}$ with $\alpha = -0.7$ (Condon 1992); flux densities are quoted in mJy throughout. Although assuming the cluster redshift for all sources introduces uncertainties for foreground and background galaxies, the same qualitative behaviour is observed in the COSMOS field where source-specific redshifts are available.

4. RESULTS

We now quantify the extent to which 1.4 GHz radio luminosity encodes information about star formation rate (SFR). Infrared-derived SFRs are treated as the reference measure of star formation activity. Our objective is not to refine a calibration, but to determine the extent to which radio luminosity can reliably recover SFR under realistic observational conditions.

4.1. Global Radio Luminosity–Star Formation Rate Relation

We begin by examining the empirical relation between rest-frame 1.4 GHz radio luminosity and infrared-derived SFR in the VLA–COSMOS sample, shown in Figure 1.

The relation is approximately linear in log–log space, but exhibits a slope shallower than unity ($m < 1$) and substantial intrinsic scatter (Yun et al. 2001; Bell 2003; Delhaize et al. 2017).

We fit a linear relation in log–log space and find a best-fit slope of $m = 0.574 \pm 0.008$ for the full sample. The Pearson correlation coefficient is $r = 0.652$, implying that $\sim 42\%$ of the variance in SFR is linearly associated with radio luminosity (i.e., $r^2 \approx 0.43$). We emphasize that Pearson r is a descriptive measure of monotonic association, whereas the R^2 values reported below are obtained from explicit predictive models evaluated on held-out data (Section 3.3).

AGN hosts exhibit a flatter relation ($m \simeq 0.27$), but they do not dominate the overall dispersion. Importantly, the majority of AGN host galaxies and radio-excess sources lie within 3σ of the best-fitting radio luminosity–SFR relation defined by the non-AGN population. These systems therefore do not constitute extreme outliers that artificially inflate the observed scatter, but instead populate the same broad distribution as star-forming galaxies. The weakness and shallow slope of the radio–SFR relation cannot therefore be attributed to a small subset of pathological AGN-dominated sources; rather, the scatter reflects intrinsic physical variability in the connection between radio emission and star formation (Lacki et al. 2010; Murphy 2009; Schleicher & Beck 2016).

A slope shallower than unity implies a luminosity-dependent bias when a proportional calibration is assumed. For $m = 0.574 \pm 0.008$, this corresponds to systematic errors of ~ 0.5 dex per decade in $L_{1.4}$ relative to the pivot luminosity: a source one decade above the pivot has its SFR overestimated by a factor of ~ 3.2 , while a source one decade below is underestimated by a similar factor. Two decades away the bias approaches ~ 0.8 – 1.0 dex. The lower panel of Figure 1 illustrates this directly, and the mathematical origin of these biases is discussed further in Section 5.2. These biases persist even after excluding AGN-classified sources, confirming they are intrinsic to the radio–SFR mapping rather than an artefact of contamination.

4.2. Machine Learning Recovery of Star Formation Rates

We next assess predictive performance using supervised machine learning models trained on radio-only inputs. Across all models, mean absolute errors exceed several hundred $M_{\odot} \text{ yr}^{-1}$, corresponding to typical uncertainties of $\gtrsim 0.5$ – 1.0 dex. The best-performing algorithms achieve $R^2 \lesssim 0.45$ in log-space, meaning that more than half of the intrinsic variance in SFR is not captured by radio luminosity alone. Across all six models, predicted values scatter broadly around the 1:1 relation, with no model recovering the high-SFR tail reliably. A notable feature shared by several models is an apparent upper cap on predicted SFR values. This cap is not the same as the variance ceiling discussed in Section 5: it arises from regression-to-the-mean. When the input feature (radio luminosity) carries limited predictive information, models converge towards predictions near the training-set mean, producing an effective ceiling on predicted values at approximately the training mean plus a small multiple of the training standard deviation. This is a direct consequence of limited information content in the predictor, and does not indicate a physical limit in the underlying SFR distribution.

To assess whether this performance is consistent with a physically reasonable noise floor, we bootstrap the expected R^2 under a perfect linear mapping with observational scatter σ_{obs} added (see Section 3.3). The bootstrap baseline

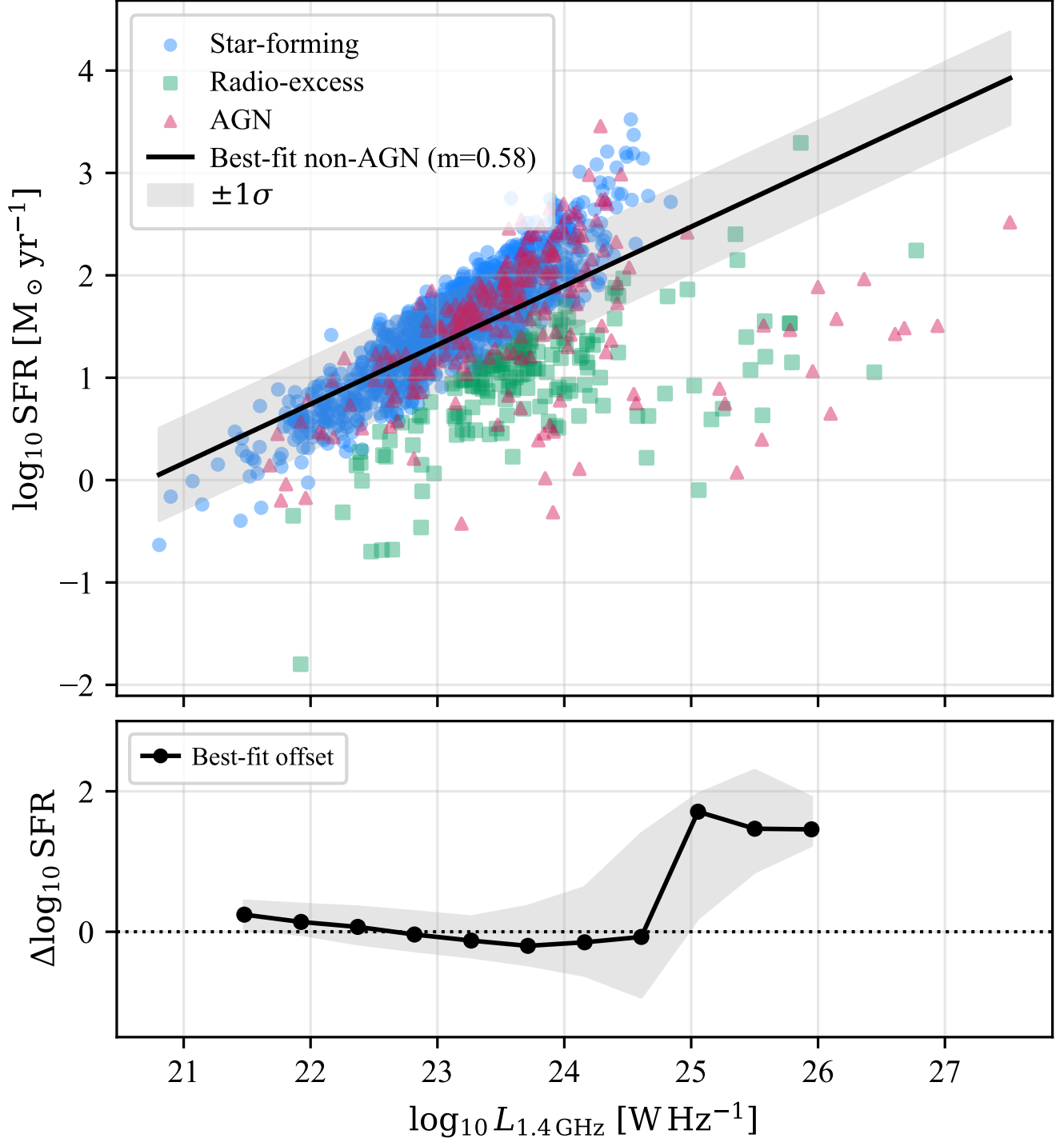


Figure 1. *Top:* Rest-frame 1.4 GHz radio luminosity versus infrared-derived SFR for the full matched COSMOS sample (training and test sets combined; $N = 7729$), separated into star-forming galaxies (blue circles), radio-excess sources (green squares), and AGN (red triangles). The solid black line shows the best-fit linear relation excluding AGN ($m = 0.58$), with the grey shaded region indicating the $\pm 1\sigma$ scatter. *Bottom:* Systematic offset $\Delta \log_{10}(\text{SFR})$ as a function of radio luminosity, binned with median $\pm 1\sigma$ shading, for the best-fit calibration. The slope of $m = 0.574 \pm 0.008$ (Pearson $r = 0.652$), significantly shallower than unity, implies systematic overestimates at high $L_{1.4}$ and underestimates at low $L_{1.4}$ for any proportional calibration, with offsets exceeding 0.3 dex at extreme luminosities.

yields $R_{\text{boot}}^2 \approx 0.85\text{--}0.90$, substantially above the $R^2 \lesssim 0.45$ achieved by all radio-only models. The gap between the bootstrap baseline and the actual model performance therefore represents the fraction of SFR variance that is genuinely unrecoverable from radio luminosity, rather than attributable to measurement noise alone.

Separating AGN and non-AGN populations (Figure 2) does not significantly improve predictive performance, confirming that the shortfall is not attributable to AGN contamination of the training set.

Table 5. Average coefficient of determination (R^2) for regression models trained using different input feature sets. Values are computed in $\log_{10}(\text{SFR})$ space and evaluated on held-out test data.

Model	Model R^2		
	1.4 GHz Only	Radio-only	Radio + WISE
Linear Regression	0.33	0.34	1.00
Support Vector Regression	0.46	0.47	0.98
k -Nearest Neighbors	0.36	0.39	0.99
Random Forest	0.20	0.27	1.00
Gradient Boosting	0.45	0.46	1.00
Multilayer Perceptron	0.44	0.47	0.99
Mean R^2	0.37	0.40	0.99

Notes: Radio-only models consistently saturate at $R^2 \lesssim 0.45$, indicating that at most $\sim 40\text{--}45\%$ of the variance in SFR can be recovered from radio luminosity alone. The near-unity R^2 values in the Radio+WISE case reflect the inclusion of infrared information directly correlated with the target variable and therefore represent a consistency check rather than an independent predictive experiment.

4.3. Environmental Dependence in MGCLS Clusters

Figure 3 shows predicted versus observed 1.4 GHz flux densities (in mJy) for both the cluster and field samples. To identify the flux density at which the predicted-to-observed flux ratio changes behaviour, marking the empirical transition from star-formation-dominated to AGN- or environment-dominated radio emission, we apply Gaussian Mixture Modelling (GMM) to the joint distribution of observed flux density and flux ratio for each cluster. GMM fits a two-component mixture to this distribution in log space, and the midpoint between the two component means defines the breakpoint. Spearman rank correlations between observed flux density and the predicted-to-observed ratio are computed in bins to quantify how correlation strength varies across the flux range. Figure 4 summarizes the resulting breakpoints and Spearman correlations across the full cluster sample.

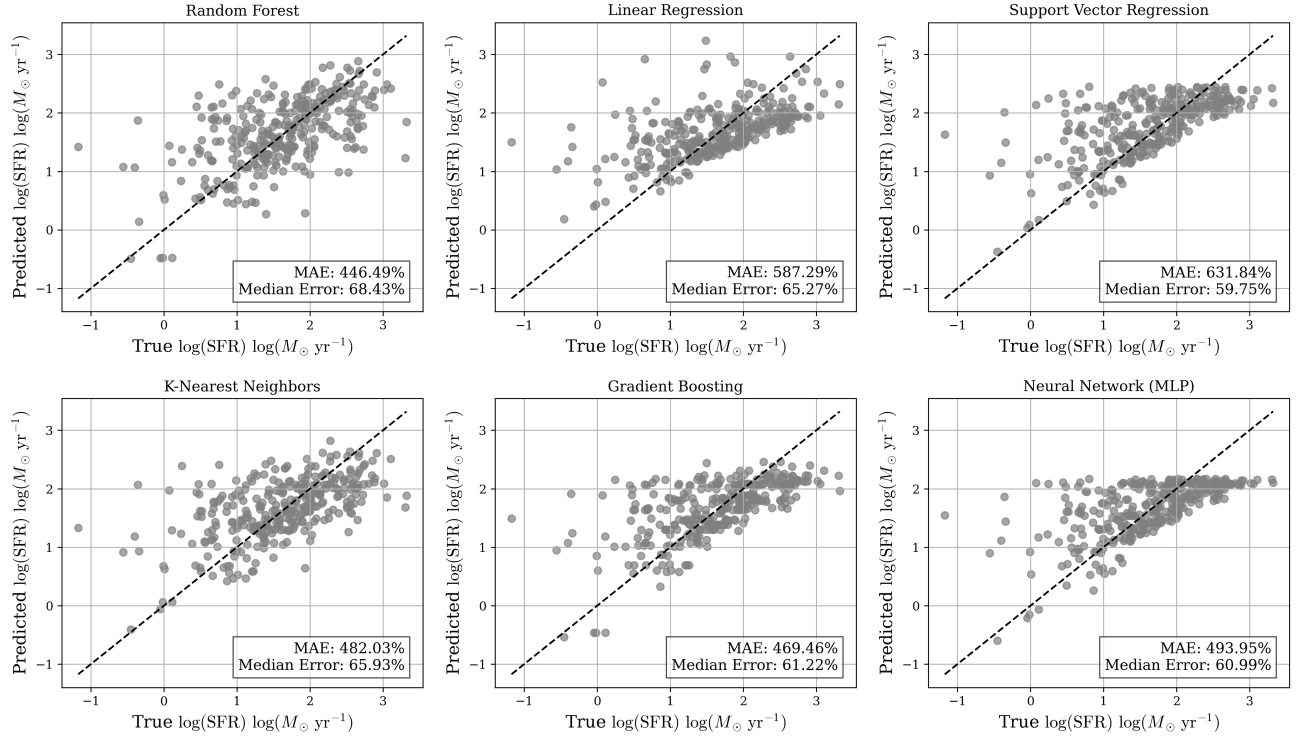
Across all clusters, no strong correlation is observed between predicted and measured flux densities. Spearman rank coefficients remain modest, particularly within R_{500} . Outside cluster cores, correlations improve modestly ($\rho \sim 0.39$), but remain weak. Within R_{500} , correlations are lower ($\rho \sim 0.21$), indicating environmental degradation (Croston et al. 2019). We emphasize that the ~ 0.2 mJy breakpoint identified via Gaussian Mixture Modelling represents an empirical transition in predictive behaviour rather than a sharp physical threshold. It marks the flux regime where additional sources of variance, such as AGN activity and cluster-related synchrotron processes, begin to dominate over star-formation-driven emission (Padovani et al. 2015; Croston et al. 2019). An illustrative example is shown in Figure 6 (Appendix A) for Abell 22, and Figure 5 shows the AGN contribution to GMM-identified populations and the AGN fraction per cluster.

4.4. Radio-excess Sources and AGN Contribution

As shown in Figure 5, moderate radio excess ($\lesssim 0.5$ dex below the mean IR–radio relation) is observed in systems without independent AGN classifications. Removing such systems does restore a substantially tighter radio–SFR relation. However, as argued in Section 2.5, the physical nature of these sources is ambiguous without resolved morphological or spectroscopic data, and their compact, smooth morphologies (Table 3) are inconsistent with a simple AGN interpretation. We therefore retain them and note that their inclusion produces a more conservative and realistic characterisation of radio-based SFR inference.

The persistence of weak correlations after masking only the independently identified AGN, without removing the broader radio-excess population, confirms that the radio-excess sources themselves carry significant predictive uncertainty, and that their classification is not straightforward from radio data alone.

Predicted vs True SFR (AGN)



Predicted vs True SFR (non-AGN)

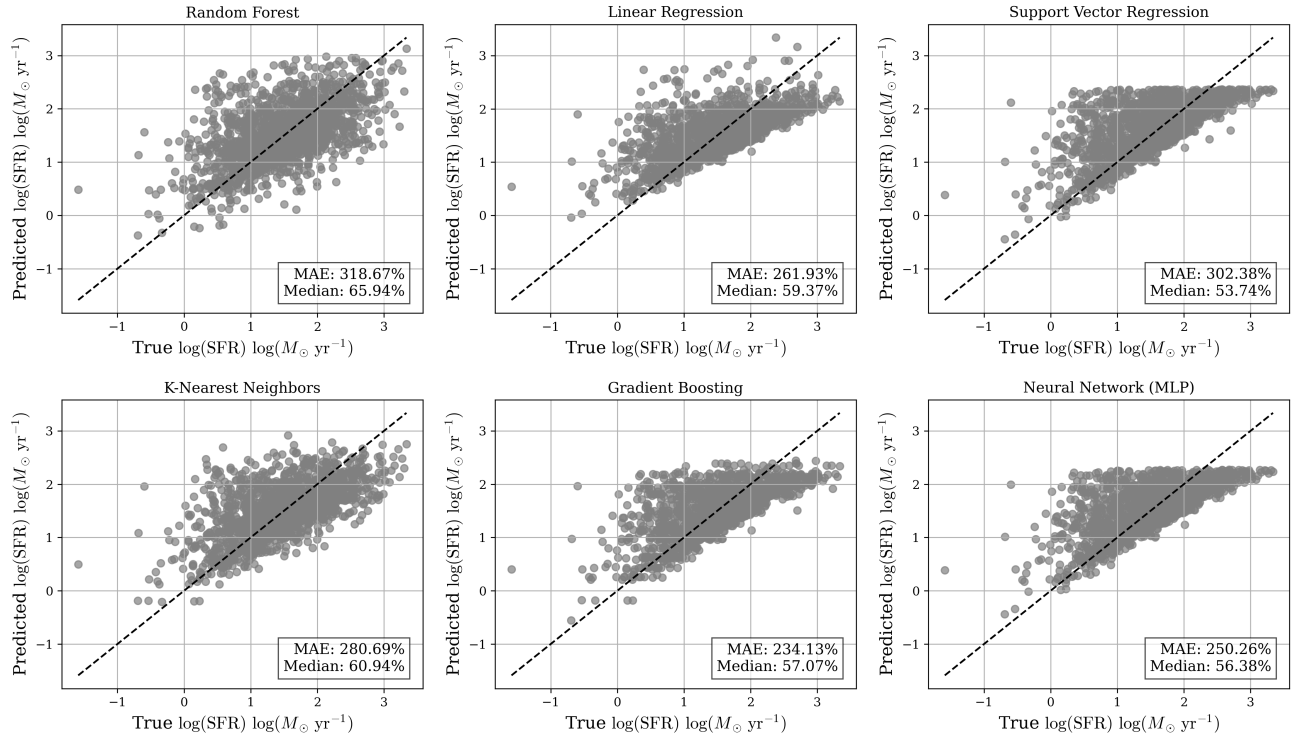


Figure 2. Predicted versus true SFRs for AGN hosts (top) and non-AGN galaxies (bottom). Significant scatter persists even after AGN removal, demonstrating that the breakdown of the radio–SFR relation is not driven solely by AGN contamination (Lacki et al. 2010; Murphy 2009).

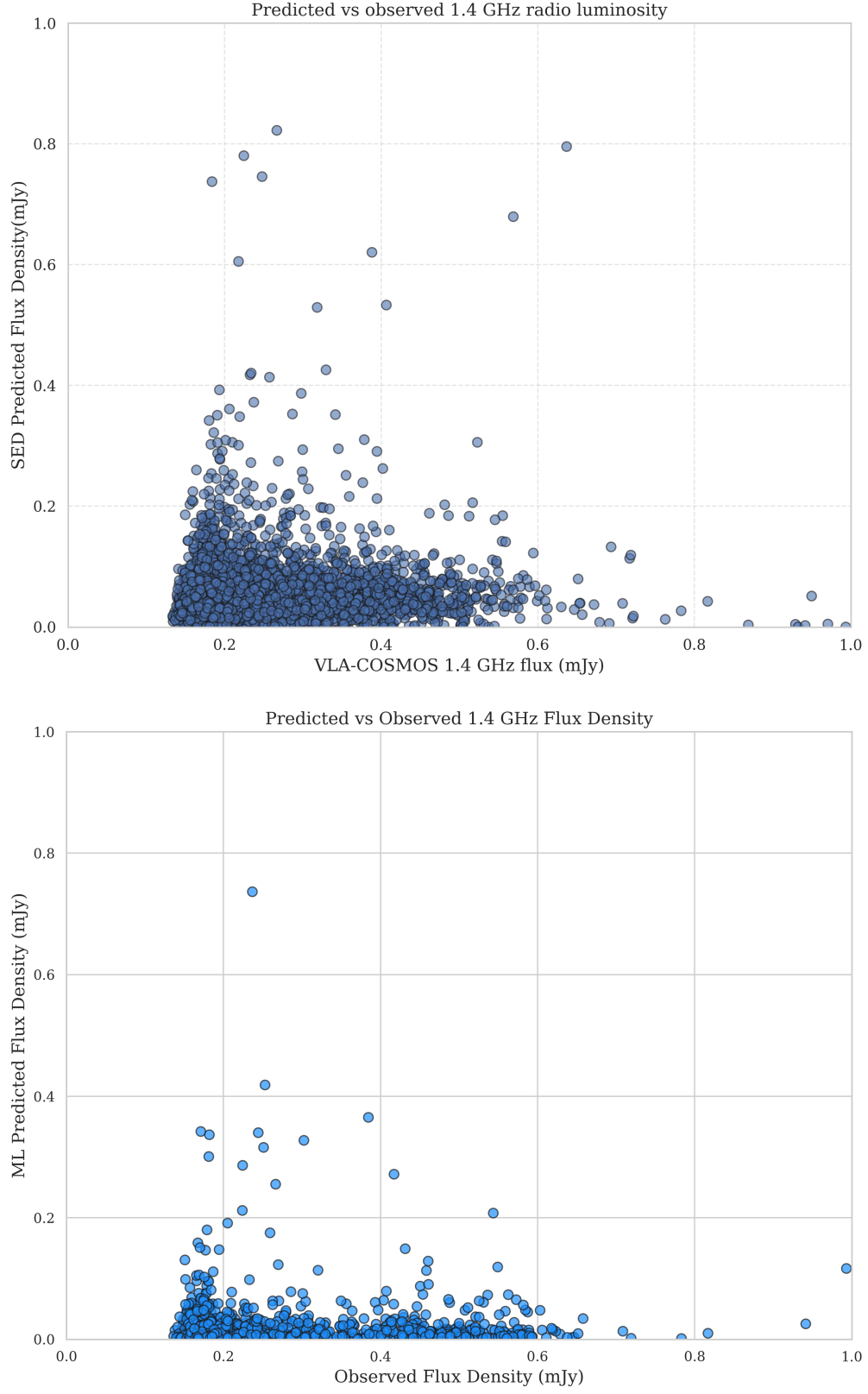


Figure 3. Comparison of predicted and observed 1.4 GHz flux densities (in mJy) in clusters (top) and the COSMOS field (bottom). No strong correlation is observed in either environment, consistent with the broad scatter in the radio–SFR relation (Yun et al. 2001; Delhaize et al. 2017).

Breakpoints Comparison (Mean vs Median)

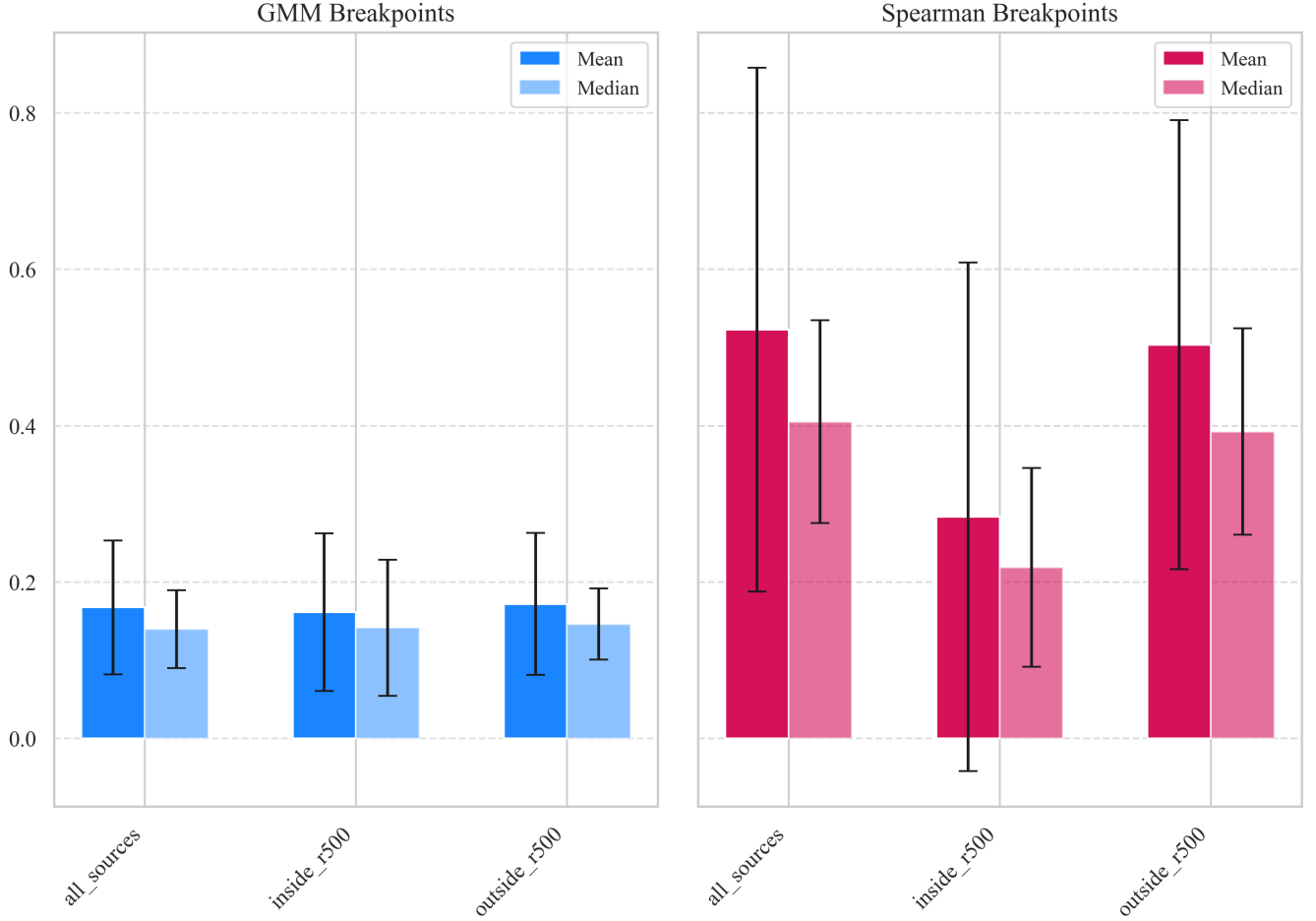


Figure 4. Mean and median Gaussian Mixture Model breakpoints and Spearman correlations for all sources, inside R_{500} , and outside R_{500} . Correlations are systematically weaker within cluster cores, consistent with environmental modulation of radio emission (Croston et al. 2019; Cuciti et al. 2021).

5. DISCUSSION

While previous studies have characterized the intrinsic scatter of the infrared–radio relation (e.g., Yun et al. 2001; Bell 2003; Delhaize et al. 2017), our analysis quantifies the maximum recoverable variance in SFR from radio luminosity alone. The convergence of predictive performance across linear and non-linear models indicates that the $\lesssim 45\%$ variance ceiling reflects the intrinsic information content of the radio features, rather than limitations of a specific regression architecture. The consistency of this ceiling across Linear Regression, Random Forest, Support Vector Regression, Gradient Boosting, and Multilayer Perceptron models demonstrates that increasing model flexibility does not recover additional predictive power, reinforcing the conclusion that the limitation is physical rather than methodological.

5.1. The Physical Origin of the Variance Ceiling

The maximum predictive performance achieved by radio-only models ($R^2 \lesssim 0.45$) implies that more than half of the intrinsic variance in SFR is not encoded in radio luminosity alone. Compared to the bootstrap baseline of $R^2_{\text{boot}} \approx 0.85\text{--}0.90$ that would be expected from a perfect linear mapping degraded only by observational scatter, the gap confirms that the shortfall is physical: radio luminosity is intrinsically decoupled from SFR by an amount that no regression architecture can recover. This variance ceiling is unlikely to arise from methodological limitations, as it persists across

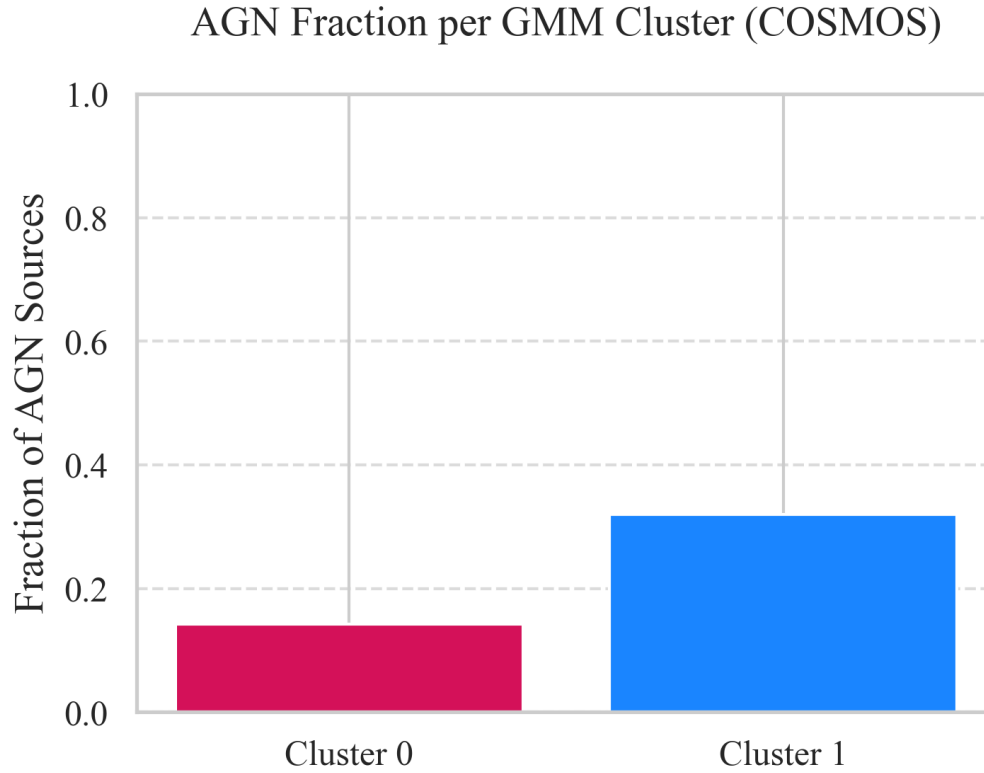
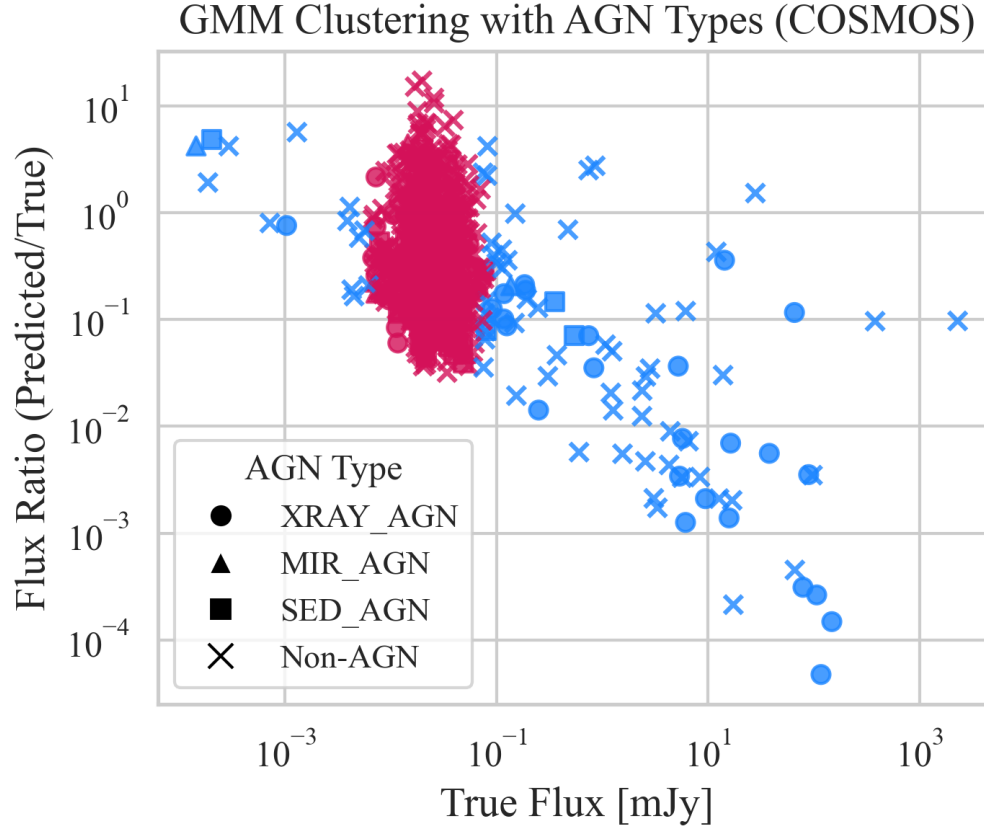


Figure 5. AGN contribution to GMM-identified populations (top) and AGN fraction per cluster (bottom). While AGN contribute to the high-luminosity tail, a substantial fraction of radio-excess sources do not carry independent AGN classifications and show morphological properties more consistent with early-type galaxies than with AGN-driven systems (Best et al. 2005; Hardcastle & Croston 2020).

linear regression, tree-based methods, kernel approaches, and neural networks. Sub-unity behaviour has previously been suggested at low radio luminosities due to cosmic-ray escape and variations in magnetic field strength (e.g., Bell 2003; Lacki et al. 2010). Our results demonstrate that this behaviour extends across environments and directly limits the recoverability of SFR from radio luminosity, even when flexible non-linear models are employed.

Several well-established mechanisms contribute to this decoupling. First, synchrotron emission depends on magnetic field strength and cosmic-ray confinement efficiency. In low-mass or diffuse systems, cosmic-ray escape can suppress synchrotron emission relative to infrared luminosity (Bell 2003; Lacki et al. 2010). Conversely, compact starbursts may approach calorimetric conditions in which relativistic electrons and positrons ($e^\pm$) lose most of their energy within the galaxy, enhancing radio emissivity per unit SFR (Lacki & Thompson 2010). Variations in magnetic field strength and gas density therefore introduce intrinsic scatter (Tabatabaei et al. 2013).

Second, radio and infrared emission respond on different timescales to changes in star formation history. Infrared emission traces dust heating by young massive stars, while non-thermal radio emission arises from supernova remnants associated with stars more massive than $\sim 8 M_\odot$ (Condon 1992; Arango-Toro et al. 2023). Rapidly varying or bursty star formation histories can therefore temporarily displace galaxies from the mean infrared–radio relation by $\lesssim 0.5$ dex without invoking AGN activity (Bressan et al. 2002; Delhaize et al. 2017).

Third, at higher redshift, inverse-Compton losses off the cosmic microwave background increasingly suppress synchrotron efficiency (Murphy 2009; Schleicher & Beck 2016), so the coupling between radio luminosity and SFR is expected to *weaken* rather than tighten with redshift. Although our cluster sample extends only to $z \sim 0.87$, the COSMOS sample reaches $z \sim 6.5$, and we observe no evidence of a restored tight relation at high redshift, consistent with this expectation (Magnelli et al. 2015; Delvecchio et al. 2021).

Finally, in cluster environments, additional radio emission from relativistic electrons in the intra-cluster medium, shocks, and turbulent re-acceleration can contaminate or enhance galaxy-scale synchrotron emission (e.g., Croston et al. 2019; Cuciti et al. 2021). The systematically weaker correlations observed within R_{500} are consistent with environmental processes introducing further variance.

Taken together, these effects imply that the radio–SFR connection is fundamentally modulated by magnetic field strength, cosmic-ray transport, environment, and star formation history. A single deterministic mapping from radio luminosity to SFR is therefore physically incomplete.

5.2. Why Linear Radio–SFR Calibrations Fail

The slope of $m \simeq 0.5$ – 0.6 , significantly shallower than the proportional mapping assumed by standard calibrations (Condon 1992; Bell 2003), implies luminosity-dependent biases of ~ 0.5 dex per decade in radio luminosity relative to any fixed pivot (Section 4.1). Proportional calibrations implicitly assume that synchrotron emission scales linearly with the rate of core-collapse supernovae, and hence with recent star formation, under conditions where cosmic-ray production and energy losses are constant across the galaxy population (Voelk 1989; Lacki et al. 2010). This bias is not removed by excluding AGN, since the shallow slope persists in the non-AGN subsample, and the majority of AGN hosts and radio-excess systems lie within 3σ of the best-fitting relation, confirming they do not dominate the scatter.

The breakdown of proportionality is therefore intrinsic rather than an artefact of extreme outliers (Lacki et al. 2010; Delhaize et al. 2017). Physically, it arises because the conditions governing synchrotron emissivity, including magnetic field strength, cosmic-ray confinement efficiency, and star formation history, vary systematically with galaxy luminosity and mass (Bell 2003; Lacki et al. 2010; Schleicher & Beck 2016), so the linear coupling between radio luminosity and SFR assumed by proportional calibrations does not hold across the luminosity range probed by deep radio surveys. Any radio-only calibration that assumes a universal linear mapping will necessarily introduce systematic, luminosity-dependent errors.

5.3. Radio-excess Sources and Realistic Survey Conditions

Removing radio-excess sources does restore a tighter radio–SFR relation: applying the standard 2σ threshold removes only 4.9% of the sample and raises r from 0.652 to 0.789, the slope from $m = 0.574$ to $m = 0.724$, and the best R^2 from 0.459 to 0.633. A stricter 1σ cut raises these to $r = 0.876$, $m = 0.811$, and $R^2 = 0.739$ (Table 4). However, this tightening is a logical artefact of the removal criterion rather than evidence of physical cleanliness. Radio-excess is defined as deviation from the infrared–radio relation; removing sources that deviate from a relation will always produce a tighter relation, regardless of why they deviate. The relevant question is not whether the relation tightens, but whether the removed sources are physically distinct from the retained population in a way that justifies their exclusion from radio-based SFR inference.

The morphological characterisation of radio-excess hosts (Table 3) demonstrates that they are not the contaminants the criterion implies. Radio-excess hosts are compact, smooth, and centrally concentrated at fixed stellar mass, redshift, and SFR, properties characteristic of passive early-type galaxies (van der Wel et al. 2014) or systems with low-luminosity AGN sustained by secular processes (Best et al. 2005; Croton et al. 2006; Hardcastle & Croston 2020), not of merger-triggered radio AGN (Holt et al. 2008; O’Dea & Saikia 2021). The flat-spectrum sub-population shows extreme asymmetry suppression ($d = -0.87$; $p = 8 \times 10^{-13}$), with a median asymmetry barely half that of matched controls, again inconsistent with the disturbed morphologies expected from major-merger-driven AGN activity. Crucially, the independently identified AGN (via X-ray, MIR, and SED criteria) are *retained* in the sample even as the radio-excess cut is applied, yet the relation still tightens substantially. This confirms that radio-excess and AGN are not the same population: the tightening is driven by the removal of sources that are offset from the mean infrared–radio relation, not by the removal of confirmed AGN.

In wide-field radio surveys, uniform deep X-ray and far-infrared data required to apply independent AGN classifications are not available (Knowles et al. 2022). Excluding radio-excess sources in such contexts imposes a selection that is neither uniformly applicable nor physically motivated, and would systematically remove a population of compact, morphologically undisturbed galaxies from the star-forming sample. Our choice to retain them reflects the realistic conditions of radio survey science and provides a more conservative and honest characterisation of the limits of radio-based SFR inference.

5.4. Implications for MeerKAT and the SKA Era

MeerKAT and forthcoming SKA surveys will probe star-forming galaxies to sub- μ Jy flux densities across wide areas and cosmic time (Knowles et al. 2022). Radio continuum measurements will therefore become increasingly central to studies of galaxy evolution. However, our results indicate that radio luminosity cannot be interpreted as a stand-alone SFR tracer without multi-wavelength context.

Even in the well-characterized COSMOS field, radio luminosity explains at most ~ 40 – 45% of the variance in SFR. In cluster environments, the predictive reliability is further degraded. A universal linear conversion from radio luminosity to SFR is therefore not supported by the data. Importantly, the qualitative behaviour observed in the COSMOS field, where SFRs are derived via full far-infrared SED fitting that explicitly models AGN contributions (Delvecchio et al. 2017; Berta et al. 2013), is reproduced in the cluster sample despite the independent WISE-based mid-infrared SFR estimator (Cluver et al. 2017). This consistency across two independent SFR methodologies argues that the breakdown of the radio–SFR relation is not driven by the choice of infrared calibration.

To quantify the practical implications for survey science, Table 6 provides empirical 68%, 95%, and 99% confidence intervals on the predicted $\log_{10}(\text{SFR})$ as a function of radio luminosity bin, derived from the model ensemble. These intervals can be incorporated directly into uncertainty budgets for radio-based SFR catalogues.

Table 6. Lookup table for predicted $\log_{10}(\text{SFR})$ as a function of $\log_{10}(L_{1.4\text{ GHz}})$. Δ_{68} , Δ_{95} , and Δ_{99} are the half-widths of the 68%, 95%, and 99% empirical confidence intervals in dex, derived from the six-model ensemble on the COSMOS test set.

$\log L_{\min}$	$\log L_{\max}$	$\log L_{\text{cen}}$	N	Median true $\log \text{SFR}$	Median pred. $\log \text{SFR}$	Δ_{68}	Δ_{95}	Δ_{99}
18.75	22.40	20.58	645	0.60	0.61	0.236	0.571	0.959
22.40	22.75	22.58	645	1.06	0.99	0.112	0.302	0.515
22.75	22.96	22.86	645	1.23	1.16	0.087	0.307	0.475
22.96	23.14	23.05	645	1.41	1.31	0.090	0.304	0.542
23.14	23.31	23.23	645	1.53	1.42	0.082	0.324	0.533
23.31	23.47	23.39	645	1.60	1.52	0.089	0.379	0.610
23.47	23.63	23.55	645	1.74	1.63	0.102	0.344	0.621
23.63	23.78	23.71	645	1.85	1.75	0.111	0.404	0.592
23.78	23.95	23.86	645	1.99	1.86	0.125	0.444	0.675
23.95	24.14	24.04	645	2.06	1.94	0.131	0.478	0.718

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$\log L_{\min}$	$\log L_{\max}$	$\log L_{\text{cen}}$	N	Median true $\log \text{SFR}$	Median pred. $\log \text{SFR}$	Δ_{68}	Δ_{95}	Δ_{99}
24.14	24.42	24.28	645	2.21	2.09	0.141	0.542	0.793
24.42	27.51	25.97	645	2.29	2.26	0.167	0.642	0.970

To understand which additional observables offer the greatest gains beyond 1.4 GHz luminosity alone, we perform an incremental information ladder analysis, adding features sequentially and recording the change in R^2 and the 68% confidence interval half-width at each step. Adding a second radio band (3 GHz luminosity) produces only a marginal improvement ($\Delta R^2 \lesssim 0.02$), consistent with the two luminosities carrying largely redundant spectral information under a fixed spectral-index assumption (Magnelli et al. 2015). Adding photometric redshift yields a modest but physically meaningful gain ($\Delta R^2 \approx 0.03\text{--}0.05$), primarily because redshift encodes the inverse-Compton loss regime and accounts for sample selection across cosmic time (Murphy 2009; Schleicher & Beck 2016). Adding stellar mass provides a similar marginal contribution. The morphological properties of radio-excess hosts documented in Table 3 (particularly the concentration, Gini coefficient, and asymmetry) encode physical conditions (gas surface density, magnetic field strength) that govern cosmic-ray calorimetric efficiency (Lacki et al. 2010; Lacki & Thompson 2010), suggesting that non-parametric morphology may represent a substantially larger information gain than any scalar radio or redshift observable. High-resolution optical imaging from Euclid, Rubin LSST, or Roman therefore represents a promising avenue for reducing radio-SFR prediction uncertainty in the SKA era.

Future radio surveys must incorporate multi-wavelength diagnostics, environmental classification, and probabilistic frameworks that explicitly account for intrinsic scatter and non-star-forming contributions. Rather than serving as a direct SFR proxy, radio continuum emission should be interpreted as one component of a multi-parameter system governed by magnetic fields, cosmic-ray physics, star formation history, and environment (Lacki et al. 2010; Murphy 2009; Schleicher & Beck 2016). Its breakdown at both high luminosities and in dense environments highlights the need for calibration strategies that are luminosity-dependent, environment-aware, and physically motivated.

6. CONCLUSIONS

We have investigated the reliability of 1.4 GHz radio continuum emission as a tracer of star formation rate (SFR) across field and cluster environments, combining deep radio observations, infrared-based SFR measurements, and supervised machine learning techniques. Our analysis spans the VLA-COSMOS field (extending to $z \sim 6.5$) and 24 galaxy clusters from the MeerKAT Galaxy Cluster Legacy Survey (MGCLS), allowing us to test the radio-SFR connection under both ideal and environmentally complex conditions.

Our principal conclusions are as follows:

- 1. Radio luminosity encodes limited information about SFR.** Even when flexible non-linear models are employed, radio-only inputs explain at most $\sim 40\text{--}45\%$ of the variance in infrared-derived SFRs ($R^2 \lesssim 0.45$). Compared to the bootstrap baseline of $R^2_{\text{boot}} \approx 0.85\text{--}0.90$ expected from a perfect linear mapping with realistic observational scatter, this gap confirms that more than half of the intrinsic scatter in SFR is physically unrecoverable from radio luminosity alone.
- 2. The radio-SFR relation has a slope significantly shallower than unity.** The best-fitting slope in log-log space is $m \simeq 0.5\text{--}0.6$, significantly shallower than the proportionality assumed by standard calibrations. A linear calibration therefore introduces systematic, luminosity-dependent biases, overestimating SFRs at high radio luminosities and underestimating them at low luminosities. The lower panel of Figure 1 shows directly that no two-parameter linear fit avoids these biases at extreme luminosities.
- 3. Radio-excess sources are morphologically distinct and cannot be straightforwardly excluded.** Removing radio-excess sources does restore a tighter radio-SFR relation, but these sources cannot be unambiguously classified as AGN from radio data alone (Padovani et al. 2015). As shown in Table 3, radio-excess hosts are significantly smoother and more centrally concentrated than matched controls, consistent with early-type or passive galaxies (van der Wel et al. 2014) rather than with disturbed AGN hosts (Holt et al. 2008; O’Dea & Saikia 2021). Their exclusion from wide-field surveys is therefore not physically justified without resolved multi-wavelength classification.

4. **Environmental effects further degrade predictive reliability.** In cluster environments, predicted and observed radio flux densities (in mJy) show no strong correspondence. Correlations are systematically weaker within R_{500} than in cluster outskirts, consistent with additional synchrotron contributions from intra-cluster processes and environmental modulation of cosmic-ray physics (Croston et al. 2019; Cuciti et al. 2021).
5. **A universal linear radio–SFR calibration is not supported.** The combination of intrinsic physical scatter, a slope shallower than unity, and environmental dependence demonstrates that 1.4 GHz radio luminosity cannot serve as a precise stand-alone SFR tracer (Lacki et al. 2010; Murphy 2009; Schleicher & Beck 2016). Empirical confidence intervals on predicted SFR as a function of luminosity are provided in Table 6.
6. **Galaxy morphology is a promising additional variable.** An incremental information ladder analysis shows that additional radio bands and redshift each contribute only marginal improvements ($\Delta R^2 \lesssim 0.05$). The structural properties of radio-excess hosts documented here (compact, concentrated, and morphologically undisturbed) encode physical conditions (magnetic field strength, gas surface density) that modulate cosmic-ray calorimetric efficiency (Lacki et al. 2010; Lacki & Thompson 2010), suggesting that high-resolution morphological information could substantially improve radio–SFR calibration. High-resolution optical imaging from Euclid, Rubin LSST, or Roman represents the most promising observational avenue in the SKA era.

Taken together, these results indicate that 1.4 GHz radio luminosity alone is insufficient for *precise* SFR inference across luminosity and environment. Radio continuum remains a valuable tracer of galaxy evolution, but its use as a quantitative SFR estimator requires probabilistic treatment of intrinsic scatter, and, where possible, multi-wavelength or environment-aware context to mitigate luminosity-dependent and environment-dependent systematics. As MeerKAT and the SKA probe star-forming galaxies to unprecedented depths, future calibration strategies must be luminosity-dependent, environment-aware, and physically motivated.

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Facilities: VLA, MeerKAT

APPENDIX

A. ABELL 22: ILLUSTRATIVE CLUSTER EXAMPLE

Figure 6 shows the predicted-to-observed flux density ratios and Spearman correlations for Abell 22, an illustrative example of the behaviour observed across the full MGCLS cluster sample. Results for all sources, sources within R_{500} , and sources outside R_{500} are shown. The pattern of weakening correlation within the cluster core is characteristic of the full sample (Figure 4).

B. DATA AVAILABILITY

The raw data used in this study are all publicly available at <https://heasarc.gsfc.nasa.gov/W3Browse/all/vlacosxoid.html>, <http://mgcls.sarao.ac.za>, and <https://vizier.cds.unistra.fr/viz-bin/VizieR?-source=II/328>.

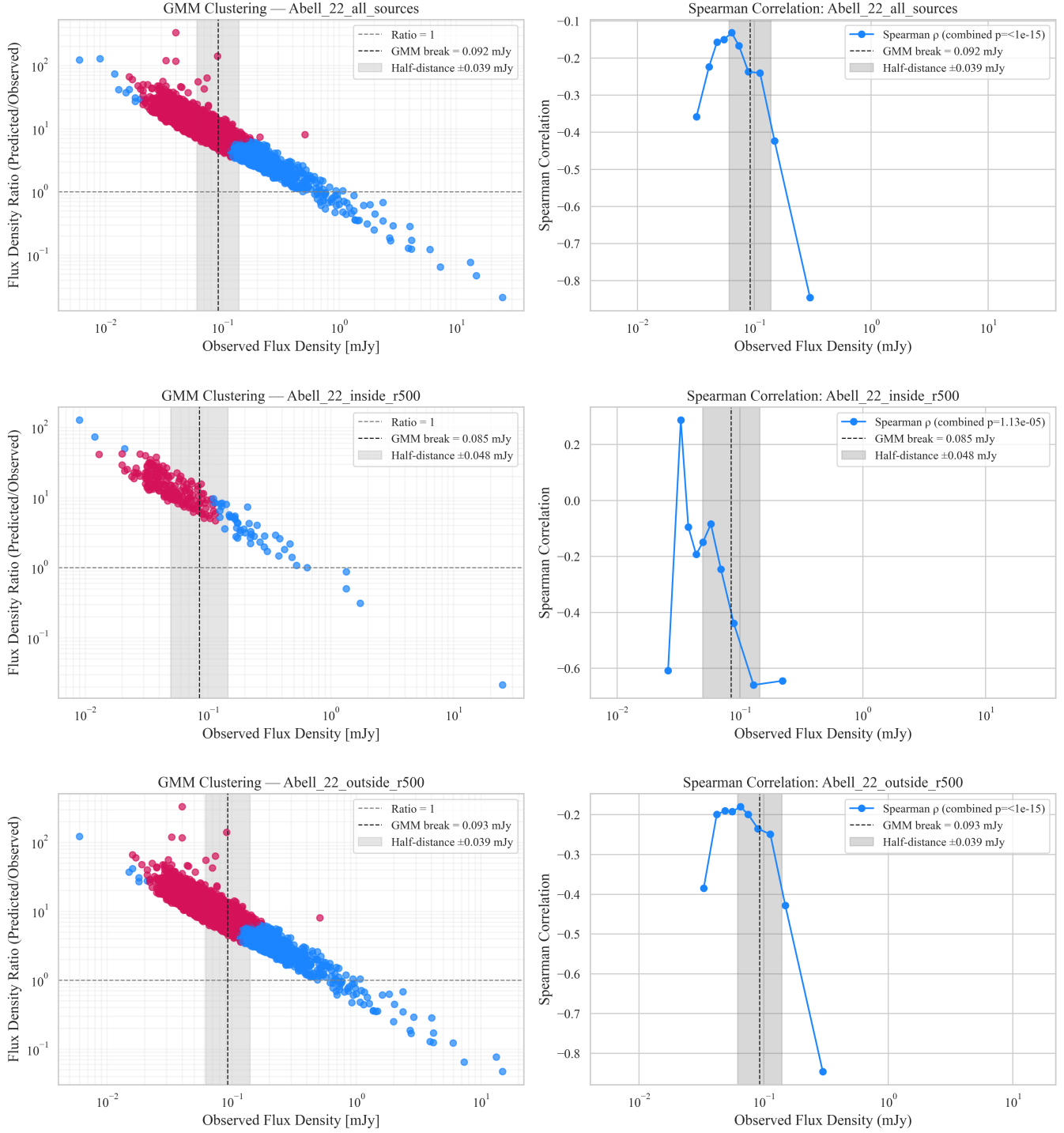


Figure 6. Predicted-to-observed flux density ratios (mJy/mJy) and Spearman correlations for Abell 22 for all sources (top), sources within R_{500} (middle), and sources outside R_{500} (bottom). Similar behaviour is observed across the full cluster sample (Figure 4).

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